

Zichen (Zachary) Wang

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SUMMARY

Computational biologist applying AI and large-scale computation to biological data for therapeutic development. Trained across wet and dry lab, bringing experimental insight to computational analysis and to the interpretation of AI models. During my Ph.D., built computational toolkits for massive 4D live-cell imaging datasets, enabling extraction of biophysical parameters and construction of whole-cell digital twins (*Cell* 2026). Ran high-content screens on the 4D platform and developed self-supervised representation learning to map cellular phenotypes across states (*Cell* 2026). Seeking to extend these approaches to new data types and disease areas as part of a cross-disciplinary team.

AREAS OF EXPERTISE

Self-supervised representation learning • Computer vision for 4D microscopy • High-content phenotypic profiling • Scalable image data pipelines • Stem-cell & organoid disease models

EDUCATION

University of California San Diego Sep. 2020 – Present
Ph.D. in Chemistry (expected Dec. 2026)

University of Illinois Urbana–Champaign Sep. 2016 – Aug. 2020
B.S. in Physics
B.S. in Molecular and Cellular Biology

RESEARCH EXPERIENCE

Graduate Researcher | 4D Cell Biology Lab, UC San Diego Mar. 2021 – Present
Johannes Schöneberg, Ph.D. | Departments of Pharmacology and Biochemistry & Molecular Biophysics

- Designed and executed multicolor 4D live-cell imaging campaigns on lattice light-sheet (LLSM) and confocal platforms, including high-content drug screens.
- Engineered iPSC-derived cell and 3D organoid models via directed differentiation and CRISPR/Cas9 editing for live-cell imaging studies.
- Developed and maintained open-source computational toolkits for segmentation, tracking, network analysis and particle-based simulation of organelles in 4D data.
- Built a GPU-accelerated, memory-efficient, OME-Zarr-based pipeline that processes terabyte-scale raw LLSM data into AI-ready datasets.
- Developed MitoSpace, a self-supervised deep-learning model that maps high-dimensional 4D cellular imaging data into latent embeddings for phenotypic screening.
- Collaborated with experimental biologists and clinicians across UCSD and external institutions on LLSM experimental design and data analysis.

Graduate Researcher | Wang Lab, UC San Diego Sep. 2020 – Feb. 2021
Wei Wang, Ph.D. | Departments of Cellular & Molecular Medicine and Biochemistry & Molecular Biophysics

- Analyzed ChIP-seq datasets to identify DNA-binding protein signatures associated with chromatin loops observed in Hi-C datasets.
- Integrated genomic databases and large-scale sequencing datasets to investigate protein sequence features linked to 3D genome organization.

Undergraduate Researcher | Kim Lab, UIUC Apr. 2019 – Sep. 2020
Sangjin Kim, Ph.D. | Department of Physics, Center for the Physics of Living Cells

- Performed single-molecule super-resolution microscopy (PALM) of protein trafficking in live bacteria.
- Developed tracking and analysis workflows for single-particle imaging datasets.
- Trained visiting students in super-resolution microscopy and laboratory techniques.

Undergraduate Researcher | Pogorelov Lab, UIUC Apr. 2018 – Sep. 2020
Taras V. Pogorelov, Ph.D. | Department of Biochemistry; Center for Biophysics & Quantitative Biology

- Ran all-atom molecular dynamics simulations on HPC clusters to study membrane–peptide interactions.

- Wrote custom scripts for molecular dynamics trajectory analysis and quantitative interpretation.
- Collaborated with experimental groups to link simulation results with biochemical and biophysical observations.

PUBLICATIONS

1. D. Agarwal*, **Z. Wang***, et al. 4D spatiotemporal landscape of mitochondrial phenotypes across cellular states unlocked through representation learning. *Cell*. 2026. DOI: 10.1016/j.cell.2026.08.028. **Equal contribution*.
2. E. Arkfeld, **Z. Wang**, et al. Whole-cell particle-based digital twin simulations from 4D lattice light-sheet microscopy data. *Cell*. 2026. DOI: 10.1016/j.cell.2026.06.010.
3. **Z. Wang**, et al. MitoTNT: Mitochondrial Temporal Network Tracking for 4D live-cell fluorescence microscopy data. *PLOS Computational Biology*. 2023. DOI: 10.1371/journal.pcbi.1011060.
4. **Z. Wang**, et al. LiveLattice: Real-time visualisation of tilted light-sheet microscopy data using a memory-efficient transformation algorithm. *Journal of Microscopy*. 2025. DOI: 10.1111/jmi.13358.
5. S. Nahar, **Z. Wang**, et al. GPU-accelerated MitoGraph for high-throughput three-dimensional mitochondrial morphology analysis. *BMC Bioinformatics*. 2026. DOI: 10.1186/s12859-026-06441-z.
6. G. McMahon, ..., **Z. Wang**, et al. 4D mitochondrial network assumes distinct and predictive phenotypes through human lung and intestinal epithelial development. *Nature Communications*. 2026. Accepted for publication.
7. M. Frei, ..., **Z. Wang**, et al. Far-red chemigenetic kinase biosensors enable multiplexed and super-resolved imaging of signaling networks. *Nature Biotechnology*. 2026. DOI: 10.1038/s41587-025-02642-8.
8. E. Tom, ..., **Z. Wang**, et al. Lipid-driven membrane remodeling engages a lysosome-dependent adaptive repair program during retinal aging. *Molecular Neurodegeneration*. 2026. DOI: 10.1186/s13024-026-00990-w.
9. P. Vo, ..., **Z. Wang**, et al. The COX2-PGE2-PKA axis suppresses antiviral immunity by inhibiting mtDNA-dependent STING activation. *Cell Reports*. 2026. Accepted for publication.
10. I. Lam, ..., **Z. Wang**, et al. Rapid iPSC inclusionopathy models shed light on formation, consequence, and molecular subtype of α -synuclein inclusions. *Neuron*. 2024. DOI: 10.1016/j.neuron.2024.06.002.
11. W. Xia, ..., **Z. Wang**, et al. Obesity causes mitochondrial fragmentation and dysfunction in white adipocytes due to RalA activation. *Nature Metabolism*. 2024. DOI: 10.1038/s42255-024-00978-0.
12. T. M. Light, ..., **Z. Wang**, et al. A cancer mutation promotes EphA4 oligomerization and signaling by altering the conformation of the SAM domain. *Journal of Biological Chemistry*. 2021. DOI: 10.1016/j.jbc.2021.100876.
13. **Z. Wang**, et al. Coaction of electrostatic and hydrophobic interactions: dynamic constraints on disordered TrkA juxtamembrane domain. *The Journal of Physical Chemistry B*. 2019. DOI: 10.1021/acs.jpcb.9b09352.
14. M. Medina-Carbonero, **Z. Wang**, et al. Mapping 4D mitochondrial phenotypes in brain organoid rosettes with lattice light-sheet microscopy. 2026. *In preparation*.

AWARDS & FELLOWSHIPS

American Heart Association <i>Predoctoral Fellowship Award No. 24PRE1196243 \$67,388 over 2 years</i> <i>Project: 4D mitochondrial profiling and drug screening for ischemia-reperfusion injury in heart cells</i>	Jan. 2024 – Jan. 2026
National Center for Supercomputing Applications <i>Director's Recognition in the Student Pushing Innovation Internship</i>	May 2020
Department of Physics UIUC <i>Philip J. and Betty M. Anthony Undergraduate Research Prize</i>	May 2020
Grainger College of Engineering UIUC <i>James Scholar Honors Program</i>	Sep. 2017 – Sep. 2018

PATENTS

1. *4D Fluorescence Microscopy Organelle Network Tracking*. U.S. Patent Application No. 18/832,021 (national phase of PCT/US2023/061312), filed Jan. 25, 2023; published as US 2025/0200763 A1, Jun. 19, 2025. Applicant: The Regents of the University of California. Patent pending.
2. *Systems and Methods for Accelerated Processing of Light-Sheet Microscopy Data*. International Patent Application No. PCT/US2025/025373, filed Apr. 18, 2025; published as WO 2025/222122 A1, Oct. 23, 2025. Applicant: The Regents of the University of California. Patent pending.

3. *A Foundation Model for Phenotypic Analysis*. International Patent Application No. PCT/US2026/015545, filed Feb. 17, 2026; published as WO 2026/174315 A1, Aug. 20, 2026. Applicant: The Regents of the University of California. Patent pending.

CONFERENCE PRESENTATIONS

- Cell Bio Meeting (ASCB/EMBO) | San Diego** Dec. 2024
"Self-supervised deep learning uncovers the semantic landscape of drug-induced latent mitochondrial phenotypes"
- Biophysical Society Annual Meeting | San Diego** Feb. 2023
"Mitochondrial Temporal Network Tracking for 4D live-cell fluorescence microscopy data"

TEACHING EXPERIENCE

- NSF-Sponsored Finding Your Inner Modeler Workshop | Chicago** Aug. 2022
Seminar on 4D organelle tracking
- Department of Chemistry and Biochemistry | UC San Diego** Sep. 2020 – Jun. 2021
Teaching Assistant: Physical Biochemistry; Protein Biochemistry Laboratory
- Center for the Physics of Living Cells | UIUC** Jul. 2019
Summer Research School

LEADERSHIP & SERVICE

- Organ-on-a-chip Symposium | UC San Diego** Nov. 2024
Organizing Committee Member

TECHNICAL SKILLS

Experimental Cell & Molecular Biology: Mammalian cell culture (immortalized/primary/human iPSC); stem cell differentiation and 3D organoid culture; molecular cloning; CRISPR/Cas9 genome editing; high-content drug-screening experiments; mitochondrial functional assays.

Advanced Fluorescence Microscopy: Lattice light-sheet microscopy; laser-scanning and spinning-disk confocal microscopy; super-resolution microscopy; multicolor 2D/3D/4D live-cell and tissue imaging.

Programming & Computational Biology: Python, C++, CUDA, MATLAB, and Bash; GPU-accelerated image processing with CuPy and CUDA; OME-Zarr for terabyte-scale image storage; Slurm/Lustre HPC workflows; segmentation and tracking of particles/cells/organelles; graph-based morphology and network analysis; particle-based whole-cell simulation; agentic coding tools for accelerated development and testing.

Deep Learning: PyTorch; scikit-learn; self-supervised representation learning; vision foundation models; dimensionality reduction and high-dimensional data visualization; quantitative model evaluation and benchmarking; biological interpretation of model predictions.

REFERENCES

Johannes Schöneberg, Associate Professor of Pharmacology and Chemistry & Biochemistry – UC San Diego
Email: jschoeneberg@health.ucsd.edu

Taras V. Pogorelov, Assistant Professor of Biochemistry – University of Illinois Urbana-Champaign
Email: pogorelo@illinois.edu